

Uptake, Translocation, Metabolism, and Distribution of Glyphosate in Nontarget Tea Plant (*Camellia sinensis* L.)Mengmeng Tong,^{†,§} Wanjun Gao,^{†,§} Weiting Jiao,[†] Jie Zhou,[†] Yeyun Li,[†] Lili He,[‡] and Ruyan Hou^{*,†}[†]State Key Laboratory of Tea Plant Biology and Utilization; International Joint Laboratory on Tea Chemistry and Health Effects, School of Tea and Food Science & Technology, Anhui Agricultural University, Hefei 230036, P. R. China[‡]Department of Food Science, University of Massachusetts, Amherst, Massachusetts 01003, United States Supporting Information

ABSTRACT: The uptake, translocation, metabolism, and distribution behavior of glyphosate in nontarget tea plant were investigated. The negative effects appeared to grown tea saplings when the nutrient solution contained glyphosate above 200 mg L⁻¹. Glyphosate was highest in the roots of the tea plant, where it was also metabolized to aminomethyl phosphonic acid (AMPA). The glyphosate and AMPA in the roots were transported through the xylem or phloem to the stems and leaves. The amount of AMPA in the entire tea plant was less than 6.0% of the amount of glyphosate. The glyphosate level in fresh tea shoots was less than that in mature leaves at each day. These results indicated that free glyphosate in the soil can be continuously absorbed by, metabolized in, and transported from the roots of the tea tree into edible leaves, and therefore, free glyphosate residues in the soil should be controlled to produce teas free of glyphosate.

KEYWORDS: glyphosate, absorption, translocation, metabolism, distribution, aminomethyl phosphonic acid

■ INTRODUCTION

Pesticides can be classified as nonsystemic or systemic based on their transportation abilities in plant tissues.¹ Systemic pesticides can penetrate plant tissues, move through the xylem or phloem, and be metabolized by the plant.² Systemic pesticides and their metabolites may have good effects on the targeted biology such as insects, fungus, or weeds. Glyphosate (PMG, N-[phosphonomethyl]-glycine), a globally sold systemic herbicide, is widely used to remove annual or perennial weeds by killing the whole plant.³ In target weeds, glyphosate can be transported from the fresh shoots to the roots and be metabolized to aminomethyl phosphonic acid (AMPA).

AMPA is a phytotoxin⁴ and genotoxic to aquatic organisms.⁵ Research on human cell lines and in mice has suggested that AMPA is genotoxic to mammalian models.⁶ Although only few scientific reports have stated that glyphosate may be carcinogenic to humans,⁷ a concern that has been received more attention in recent years. To ensure food safety, many countries and organizations have set maximum residue limits (MRLs) for glyphosate in agriculture products. In commercial tea (made from leaves of *Camellia sinensis* L.), the MRLs for glyphosate are 2.0 mg kg⁻¹ in the European Union and 1.0 mg L⁻¹ in Japan and China.⁸

The uptake, transport, and metabolism of glyphosate have been studied in target plants during the thorough investigation of the mechanism by which it inhibits different weeds.^{9–12} Glyphosate can be a pollutant to nontarget plants through direct contact or through release from soil, where it can be absorbed by the roots and then transported to other parts of the plant.¹³ When glyphosate sprayed in soil, 4% of it can be extracted.¹⁴ Glyphosate persisted in soil for 30 to 60 days after applications of 0.5 to 2.0 kg ha⁻¹ in tea crops and it residues in the tea leaves were detected up to 15 days at all three treatment doses.¹⁵ There has been some research about the metabolism of

glyphosate into AMPA in herbaceous weed species.^{9,16} However, there are few reports concerning the uptake and transport of PMG/AMPA in woody plants. ¹⁴C-glyphosate was studied in the leed tree, where the metabolites of AMPA, sarcosine, and other unknown components were detected.¹⁷

As a perennial woody plant, the tea tree is one of the most important economic crops in China. Glyphosate is an effective herbicide used in tea plantations to control weeds around the roots of tea trees. Glyphosate residues have been found by UPLC–MS/MS in 18.6% of a sampling of tea products¹⁸ at 0.105–3.223 mg kg⁻¹. However, there is no published report about the capabilities of the tea tree to uptake, transport, or metabolize glyphosate or about the distribution of glyphosate or its metabolite AMPA within the tea tree. It is relatively difficult to detect glyphosate and AMPA using conventional methods because they are highly polar, nonvolatile, and lack chromogenic and fluorescent groups. Gas chromatography (GC),^{19–21} gas chromatography/mass spectrometry (GC/MS),²² high-performance liquid chromatography (HPLC),^{23–25} and liquid chromatography–mass spectrometry (LC–MS)^{26–31} are the common used methods for determining glyphosate and AMPA. The best selectivity and sensitivity are usually achieved using an HPLC–MS/MS method based on precolumn derivatization with 9-fluorenylmethyl chloroformate (FMOC-Cl).³² This combined method has gradually come to play an important role in the analysis of glyphosate and AMPA. LC–MS/MS-based methods using solid phase extraction (SPE) cleanup coupled with derivatization by FMOC-Cl are available for determination of glyphosate and AMPA levels

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from tea samples.¹⁸ While the time-consuming and costly SPE cleanup step may improve method sensitivity, it may also increase the signal variability. The use of a QuEChERS (quick, easy, cheap, effective, rugged, and safe) extraction approach has been developed with a pretreatment method for the analysis of multiple pesticides in food. GCB, an absorbent of pigments and sterols, was commonly used in the QuEChERS method. Recently, an inexpensive and excellent absorbent, polyvinylpolypyrrolidone (PVPP), has been shown to eliminate the abundant and interfering polyphenols from tea matrices.³³

The aim of the present study was to develop a simple, selective, and reliable method based on a QuEChERS dispersive cleanup approach and derivitization coupled with UPLC–MS/MS for the determination of glyphosate and AMPA levels in different parts of the tea plant. Furthermore, this developed method was used to study the uptake, transport, metabolism, and distribution of glyphosate in tea plants. These data reveal the interaction between glyphosate and the tea plant and provide some information about both the source and mitigation of glyphosate contamination of tea products. Knowing the distribution of glyphosate and AMPA in different parts of the tea plant provides information that can aid decision making regarding safety of tea pre- and postharvest.

MATERIALS AND METHODS

Chemicals and Reagents. Chromatography-grade acetonitrile and dichloromethane (CH_2Cl_2) were obtained from Tedia Company, OH, USA. Water used for LC–MS/MS was produced in the laboratory with a Milli-Q water purification system (Millipore, Bedford, MA). Graphitized Carbon Black (GCB, 120/400 Mesh) and C_{18} (230–400 Mesh, 60 Å; SiliCycle, Canada) were obtained from ANPEL Scientific Instrument Co., Ltd. (Shanghai, China), and polyvinylpolypyrrolidone (PVPP) was purchased from Solarbio Science and Technology Co., Ltd. (Beijing, China). An Oasis HLB cartridge for SPE (Oasis HLB, 3 mL/60 mg) was obtained from Waters Corporation (Milford, MA). KOH, acetone, sodium tetraborate decahydrate, and ammonium acetate were purchased from Sinopharm Chemical Reagent Co., Ltd. (Shanghai, China). HCl was purchased from Shanghai SuYi Chemical Reagent Co., Ltd. (Shanghai, China). The FMOC-Cl was purchased from Alfa Aesar (Tianjin, China). Formic acid was obtained from Aladdin Industrial Corporation (Shanghai, China). Glyphosate (98.0%), glufosinate ammonium (97.5%), and aminomethylphosphonic acid (99.0%) were received from Dr. Ehrenstorfer (Augsburg, Germany). Glyphosate isopropylammonium salt (41%) was obtained from Anhui Sanonda Biological Technology Co., Ltd. (Anhui, China).

Standard stock solutions of glyphosate (PMG) and aminomethylphosphonic acid (AMPA) were prepared by weighing 10 mg of each analyte and dissolving in 10 mL of water. Working standard solutions were prepared by diluting the standard stock solution with water. All solutions were stored at $-4\text{ }^\circ\text{C}$.

FMOC-Cl was dissolved in acetone at concentrations of 0.5, 1.0, 10, 20, and 40 g L^{-1} . Borate buffer consisted of 5 g of $\text{Na}_2\text{B}_4\text{O}_7 \cdot 10\text{ H}_2\text{O}$ dissolved in 100 mL of water with the pH adjusted to 9 using 5 mol L^{-1} HCl.

Annual cuttings of *Camellia sinensis* cultivar Shu Cha Zao (Shucheng County, Anhui province, China) were cultured for six months in an automated hydroponic system (Anhui Agricultural University, Hefei, China). The nutrient solution contained (in mg L^{-1}) 30 NH_4^+ , 10 NO_3^- , 3.1 PO_4^{3-} , 40 K^+ , 20 Ca^{2+} , 25 Mg^{2+} , 0.35 Fe^{2+} , 0.1 B^{3+} , 1.0 Mn^{2+} , 0.1 Zn^{2+} , 0.025 Cu^{2+} , 0.05 Mo^+ , and 10 Al^{3+} . All tea saplings displayed the same growth rate and were 15–20 cm in height. All cultivation experiments were done in the greenhouse at Anhui Agricultural University.

LC–MS/MS Analysis. *Tea Sapling Sample Preparation.* About 5 g of leaves, stems, and roots from tea plants was picked, cut into pieces, and mixed homogeneity, and then a 0.25 g aliquot of the

samples was put into mortar, to which was added 10 mL of water, and then ground. After that, the roots or stem samples mixture was sonicated for 10 min and leaves sonicated for 30 min, respectively. After ultrasonic extraction, the samples were centrifuged at 5000 rpm for 5 min. The aqueous supernatant was transferred to a new centrifuge tube, mixed with 2.5 mL of CH_2Cl_2 by vortex for 2 min, and centrifuged at 5000 rpm for 5 min. A 2 mL aliquot of supernatant was transferred into a 5 mL centrifuge tube to which 5 mg of GCB and 50 mg of PVPP were added. The mixture was shaken by vortex for 2 min and then centrifuged at 10 000 rpm for 10 min. The supernatant (1.0 mL) was transferred into a 5 mL centrifuge tube and mixed with 1 mL of borate buffer by vortex for 2 min. FMOC-Cl (1.0 mL of 20 g L^{-1}) was added to the mixture and allowed to react overnight at room temperature. The reaction solution was filtered through a $0.22\text{-}\mu\text{m}$, hydrophilic PTFE needle filter for subsequent UPLC–MS/MS analysis.

LC–MS/MS Analysis. The LC–MS/MS system included an Agilent Series 1290 ultraperformance liquid chromatography system (UPLC) and an Agilent 6460 triple quadrupole mass spectrometer (QQQ; Agilent Technologies, Palo Alto, CA, USA). The UPLC system was equipped with a quaternary pump, a vacuum solvent degasser, a column oven, and an autosampler. A Waters HSS T3 column (particle size, $1.8\text{ }\mu\text{m}$; length, 100 mm; internal diameter, 2.1 mm) was used with a solvent flow rate of 0.3 mL min^{-1} . The column compartment temperature was set at $40\text{ }^\circ\text{C}$, and the injection volume was set at $5\text{ }\mu\text{L}$. Mobile phase A was 0.1% formic acid in water containing 2 mmol L^{-1} ammonium acetate and B was 0.1% formic acid in acetonitrile. The solvent gradient was as follows: 0–0.5 min 5% B, 0.5–6 min 50% B, 6–7 min 95% B, 7–9 min 5% B, and 9–14 min 5% B, according to the Chinese standard method SN/T 1923–2007.³⁴

The mass spectra were acquired using electrospray ionization (ESI) in the positive ionization mode. Analyses of glyphosate and AMPA were performed in multiple reaction monitoring (MRM) mode. The settings were: a drying gas flow of 6 L min^{-1} with a drying gas temperature of $325\text{ }^\circ\text{C}$, a nebulizer pressure of 45 psi, a sheath gas temp of $350\text{ }^\circ\text{C}$, and a sheath gas flow of 11.0 L min^{-1} . The fragmentor voltage for PMG-FMOC and AMPA-FMOC was 135 V, cell accelerator voltage was 7 V, and collision energies were 11 and 15 eV, respectively. The mass transition ion-pair of PMG-FMOC and AMPA-FMOC were $m/z\ 392 \rightarrow 88$ and $334 \rightarrow 179$, respectively.

Validation of Analytical Procedure. Evaluation of the method included fitting to linear equations and determining the matrix effect, recovery rate, and limit of quantification (LOQ). The standards produced a linear result between 5 and $500\text{ }\mu\text{g L}^{-1}$. The matrix effect (ME), the change of ionization efficiency in the presence of other compounds, was expressed as the responses of FMOC derivatives of PMG and AMPA in matrix compared to the signal in solvent, calculated by the following equation:³⁵

$$\text{ME (\%)} = \left(\frac{\text{Peak area (spiked extract)}}{\text{Peak area (solvent standard)}} - 1 \right) \times 100$$

An ME value equal to 0% means that no matrix effect was detected, while positive and negative values indicate enhancements and suppressions, respectively, of the analyte signal by matrix compounds. Matrix effects were classified into different categories based on the value of this percentage. The matrix effect was not obvious when the values of was within $\pm 20\%$.³⁶ The matrix effect in this proposed method was evaluated in fresh tea leaves spiked with 0.5 mg kg^{-1} compared with the same concentration of standard sample. Leaves were used because they are a more complex matrix than roots or stems. The recoveries of glyphosate and AMPA in roots, stems, and leaves at spiked levels of 0.5 or 2 mg kg^{-1} using standard calibration, each concentration level repeated six times. The limit of quantitation (LOQ) was calculated as a signal-to-noise ratio of 10 ($\text{S/N} = 10$) using the lowest responding concentration for each pesticide at the primary ion transition (quantitation ion transition) obtained from the MS/MS mode.

Uptake, Transport, Metabolism, and Distribution of Glyphosate in Tea Saplings. *Phytotoxicity of Glyphosate to Tea*

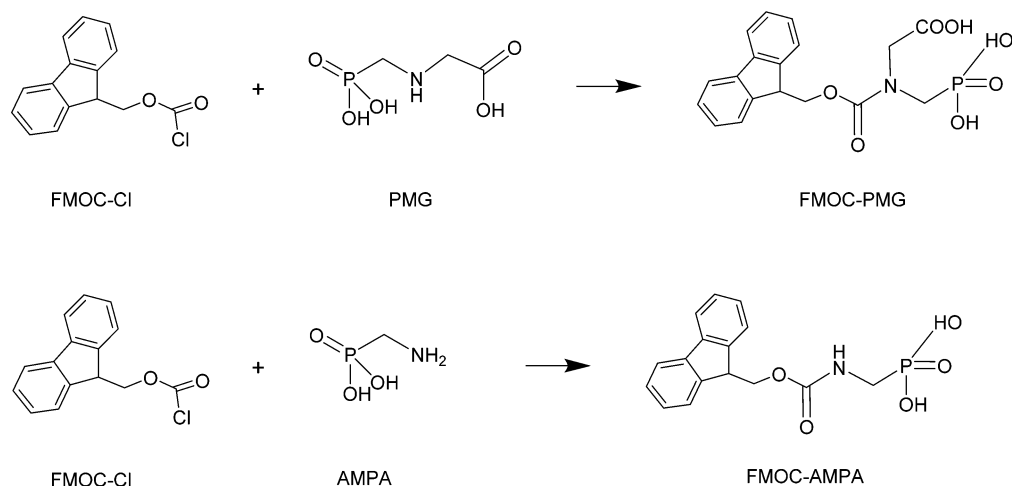


Figure 1. Derivatization of glyphosate (PMG) and its metabolite AMPA with 9-fluorenylmethyl chloroformate (FMOC-Cl).

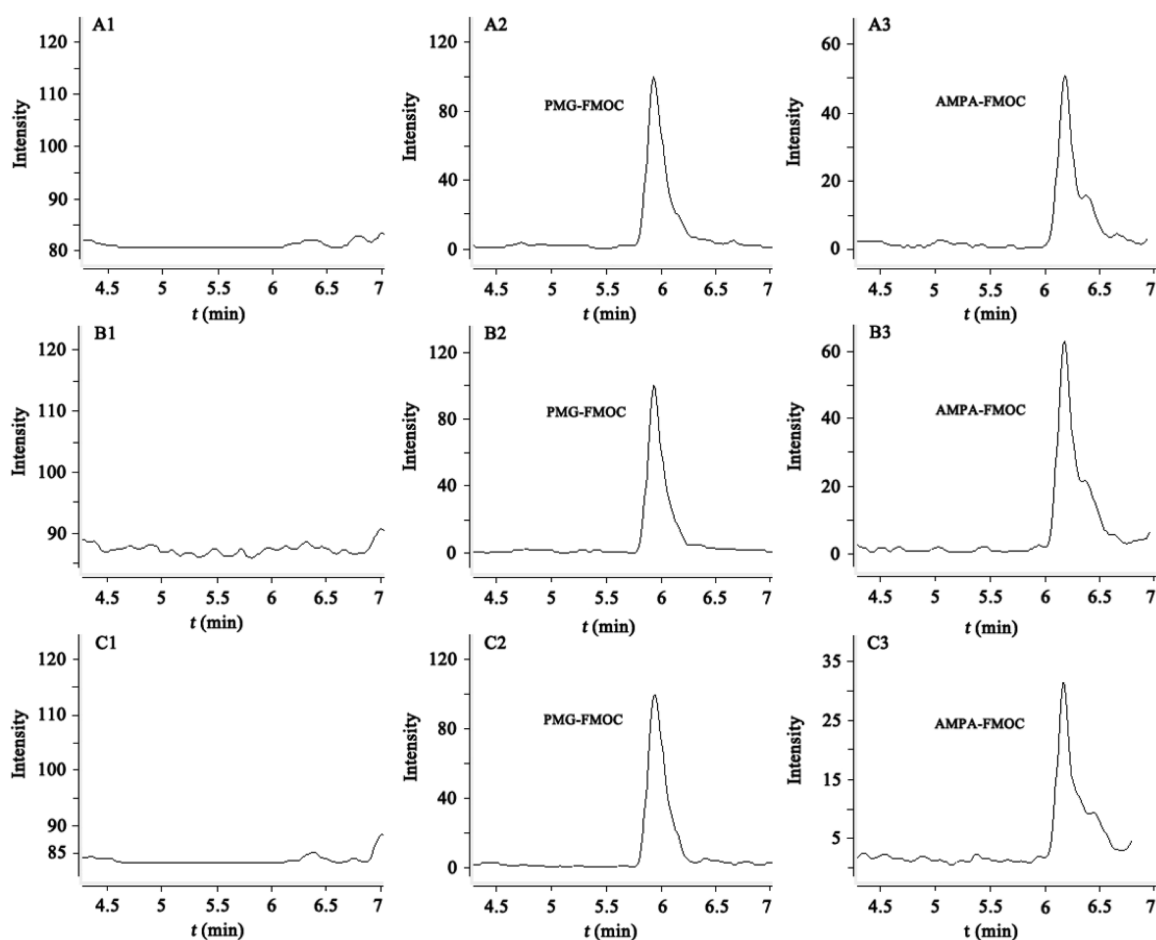


Figure 2. MS/MS total ion chromatograms of derived compounds in blank matrix (first column) and from (A) leaves, (B) stems, or (C) roots spiked (0.5 mg kg^{-1}) with PMG (second column) or AMPA (third column).

Saplings. Tea saplings were selected from the hydroponic system and moved into one of five red plastic buckets. Five tea saplings were cultivated in 1.2 L of nutrient solution containing 0, 5, 50, 200, or 2000 mg L^{-1} glyphosate (41% glyphosate isopropylammonium salt) in different buckets. After 1, 3, 5, 7, 10, and 14 days, growth, wilting, and phytotoxicity of the tea saplings were noted.

Uptake, Transport, Metabolism, and Distribution of Glyphosate. Tea saplings were transferred from the hydroponic system into blue plastic buckets, with 30 tea saplings cultivated in 6 L of nutrient

solution containing 5 mg L^{-1} glyphosate. After 0, 1, 3, 5, 7, 10, 14, and 21 days, different parts of the saplings or the whole sapling were collected for determination of the content of glyphosate and AMPA.

RESULTS AND DISCUSSION

Sample Preparation. Derivatization Reaction. The derivatization reaction is shown in Figure 1. Among the tree published methods for derivatization of PMG and AMPA, the concentration of FMOC-Cl in solvent differed significantly.

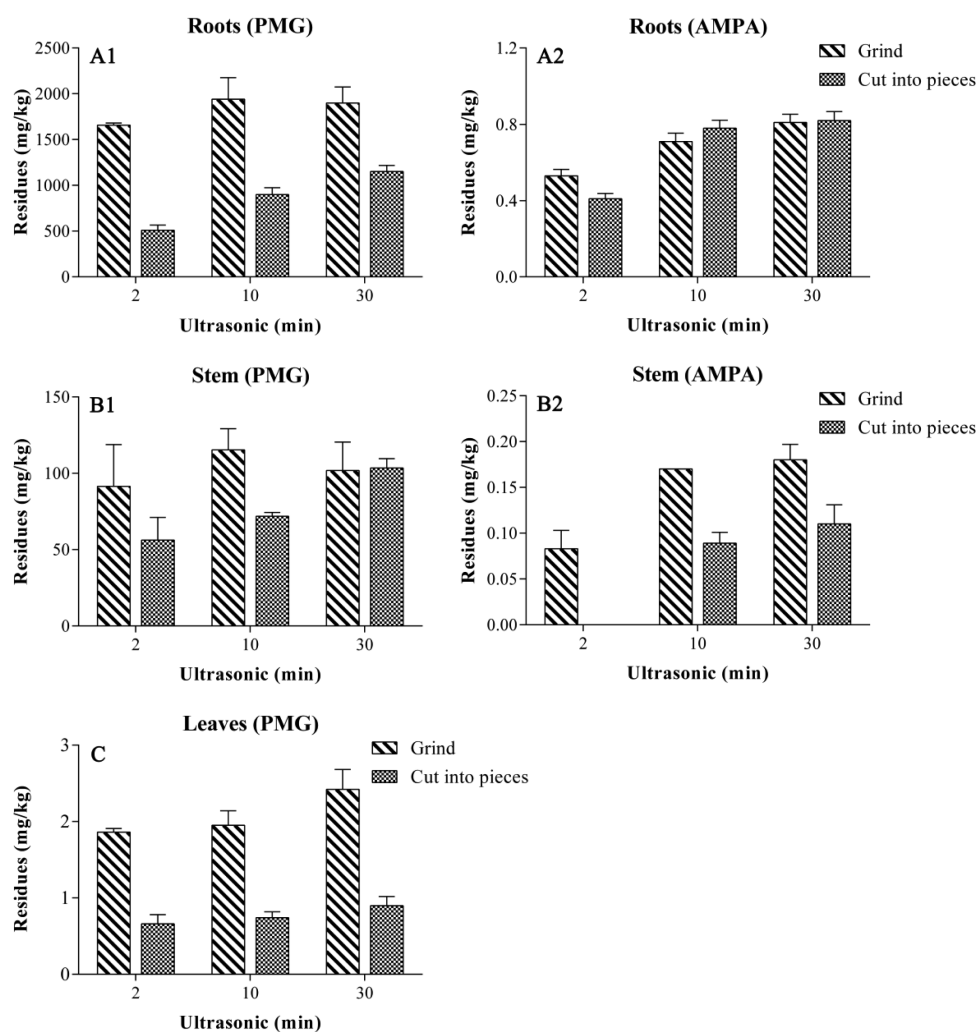


Figure 3. Extraction contents of PMG and AMPA of different part of tea plant by two extraction methods of cutting and grinding. (A1, A2) Roots; (B1, B2) stem; (C) leaves.

One study used 0.1 mL of 10 g L⁻¹ FMOC-Cl for derivatization,¹⁸ another used 1.0 mL of 20 g L⁻¹ FMOC-Cl,²⁷ and the latest research used 0.3 mL of 1.5 g L⁻¹ FMOC-Cl.³⁷ In this study, different concentrations (1.0 mL) of the derivation reagent FMOC-Cl (0.5, 1.0, 10, 20, or 40 g L⁻¹) were mixed with 0.05 mg L⁻¹ of the pesticide standards. The average peak areas of glyphosate were 188.7, 214.7, 186.3, 212, and 205.7 and of AMPA were 202, 234.3, 220.7, 231.7, and 239.7 with the different concentrations of FMOC-Cl. Perhaps 1.0 g L⁻¹ is the economical choice, but 20 g L⁻¹ slightly increased the response of them in tea samples compared with the standard samples. Thus, 20 g L⁻¹ FMOC-Cl was used in our proposed method.

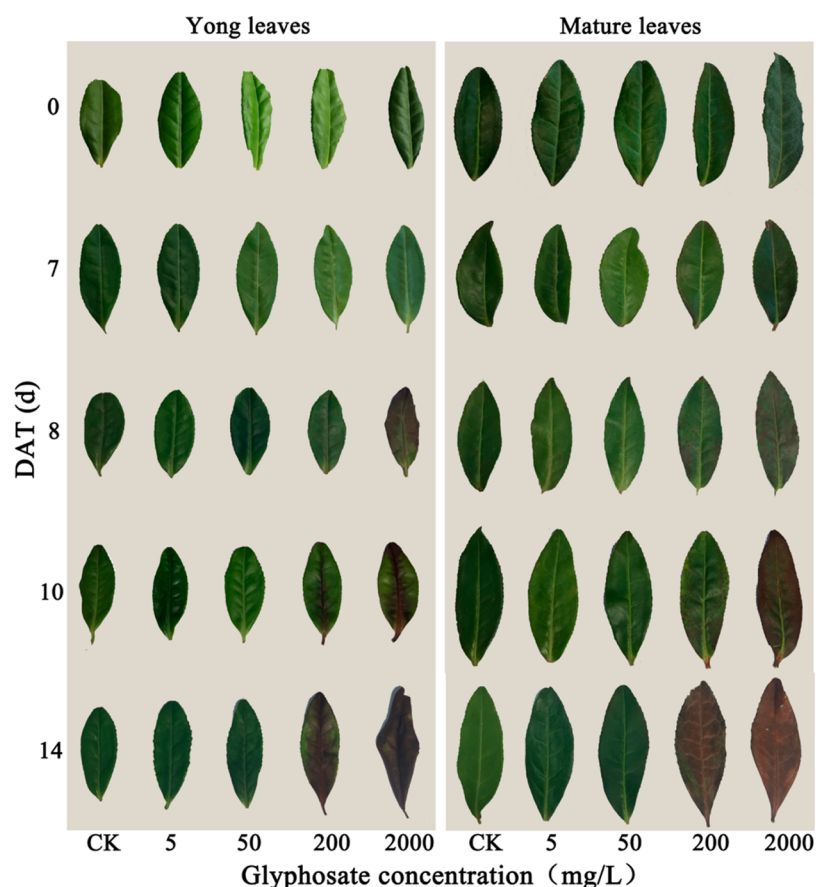
Extraction. Glyphosate and AMPA are strongly polar, water-soluble compounds. Water has been used as the extraction solvent in most published methods, whereas different extract methods and treatment times were used, such as sonication or grinding. In this study, glyphosate-treated samples (about 3–5 g) were cut to small pieces. A 0.25 g aliquot of the cut pieces of sample was ground in mortar and then sonicated for 2, 10, or 30 min to compare the extraction amounts for glyphosate and AMPA. The MS/MS total ion chromatograms for the derived compounds in blank matrix and 0.5 mg kg⁻¹ spiked samples were shown in Figure 2. The results showed that there is a

baseline separation of glyphosate and AMPA and little interference in the trace of AMPA. The extraction amount of different method was shown in Figure 3. The amounts of glyphosate and AMPA recovered were significantly lower with the cutting method than with the grinding method. A longer sonication time of 10 min increased the extraction of glyphosate in all samples. However, sonication for longer than 10 min caused a slight decrease in the extraction of PMG from roots and stems, although extraction from leaves still increased with increase sonication time. There were no significant differences in extraction of APMA from roots or stems when sonication increased from 10 to 30 min. From these preliminary extraction tests, the optimized extraction used in the study consisted of grinding all samples and extracting root and stem samples with sonication for 10 min and the fresh leaf samples for 30 min.

Cleanup Method. Strategies used to minimize matrix interference include improvement of chromatographic selectivity, to avoid interference of coextracted matrix components, and modification of sample preparation. Tea represents a complex matrix, containing high amounts of free amino acid (1–2%), polyphenols (18–36%), and alkaloids (2–4%), which can easily be coextracted with target pesticides and may interfere with the subsequent derivatization reaction, especially the free amino

Table 1. Average Recovery Rates, Relative Standard Deviation (RSD), and Limit of Quantification (LOQ) of PMG and AMPA Compounds from Leaves, Stems, and Roots ($n = 6$)

		spiked level				
		0.5 mg/kg		2 mg/kg		
matrix	compound	recovery (%)	RSD (% , $n = 6$)	recovery (%)	RSD (% , $n = 6$)	LOQ (mg/kg)
leaves	PMG	84.2	5.87	82.3	10.82	0.1
	AMPA	72.3	10.14	94.6	13.06	0.1
stem	PMG	91.7	5.95	84.9	4.70	0.05
	AMPA	91.8	6.50	93.2	8.68	0.05
roots	PMG	116.0	12.99	96.7	7.34	0.05
	AMPA	74.8	13.43	85.8	8.73	0.05

**Figure 4.** Visual phytotoxicity on tea leaves of different ages, young leaves (leaves 1–3) or mature leaves (leaves 4–6) at different days after treatment (DAT) with different concentrations of glyphosate (0, 5, 50, 200, or 2000 mg L⁻¹) delivered in the hydroponic nutrient solution.

acids.³⁸ A method has used liquid extraction with CH₂Cl₂ as solvent combined C18 column (alkylsilane bonded to silica gel) to minimize matrix disturbance during determination of glyphosate and AMPA in commercial tea.³⁶ The absorbents in HLB cartridges (*m*-divinylbenzene and *N*-vinylpyrrolidone copolymer) have similar characteristics to C18. For example, glyphosate and AMPA residues in tea were determined by alkaline solution extraction and HLB column purification.¹⁸ The difference between these methods was whether the first water extraction was followed by a re-extraction containing or lacking CH₂Cl₂. Other studies have shown that the purification effect of CH₂Cl₂ is better for fresh agricultural products extraction such as soybean,²⁶ rice, maize and soybean,³⁰ olive, and other plant materials.³⁹ Until now, there is no report about the fresh tea plant. Some coextracted unknown compounds may greatly affect the pesticide derivatization that follows. Our

recent research showed that a QuEChERS extraction method using PVPP combined with GCB effectively and efficiently cleans up tea samples for detection of pesticide residues.⁴⁰ To find a suitable cleanup method for fresh samples taken from different parts of the tea plant, three preparation methods were compared, as follows: (A) cleanup the aqueous extract with HLB; (B) re-extract the aqueous extract with CH₂Cl₂ as solvent and then cleanup with HLB cartridge; and (C) re-extract the aqueous extract with CH₂Cl₂ as solvent and then cleanup with PVPP and GCB as sorbents. As the most complex matrix of our three tissues, tea leaf was chosen to verify the effectiveness of these three cleanup methods. The pesticide standard (4 mg kg⁻¹) was added to the fresh leaf extract after aqueous solution extraction and sonication for 30 min. This mixture was cleaned up by three methods, derivatized with FMOC-Cl, and analyzed by UPLC–MS/MS. To quickly compare the recoveries

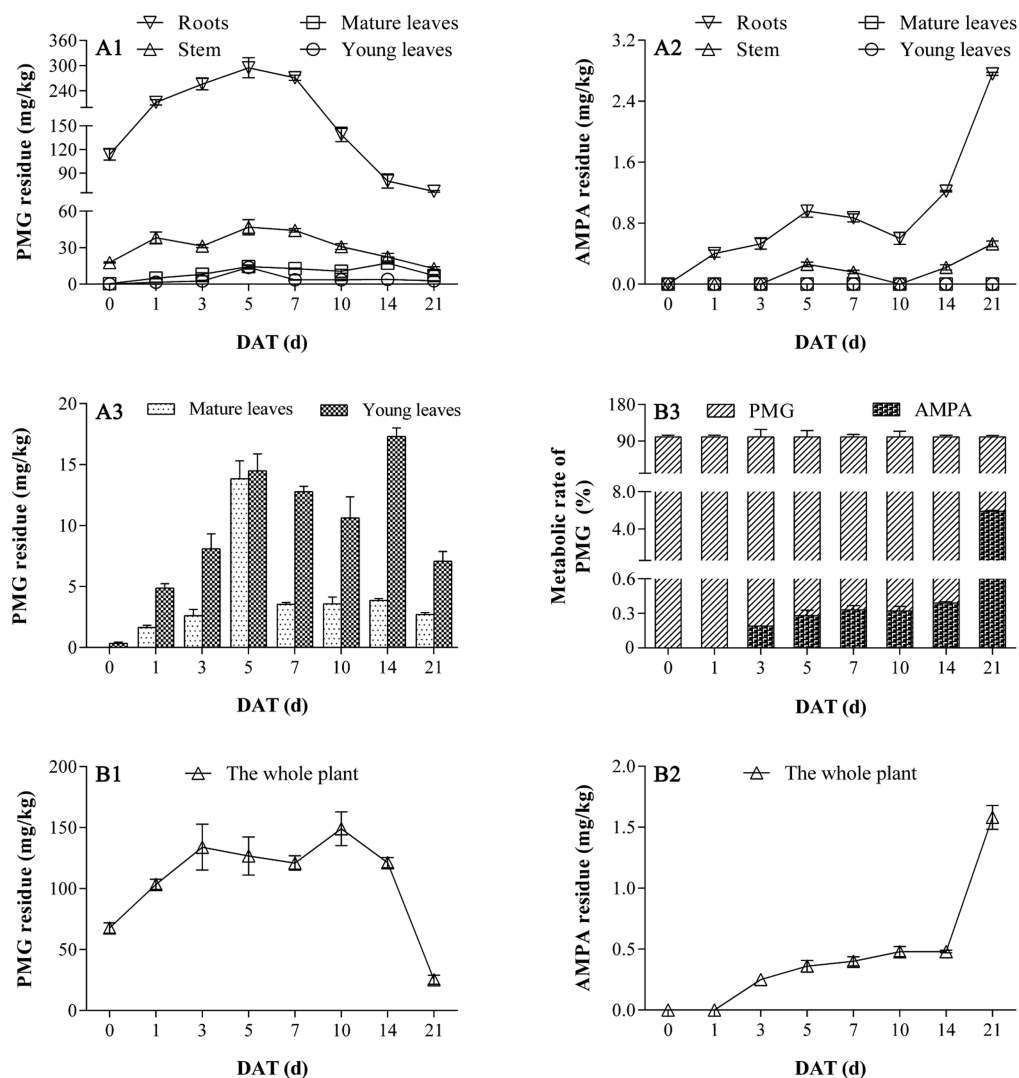


Figure 5. Concentration of PMG and AMPA in different parts of tea saplings (A1, A2) and the whole plant (B1, B2) with sustained treatment of 5 mg L⁻¹ glyphosate in nutrient solution. Comparison of the content of glyphosate in young leaves (leaves 1–3) and mature leaves (leaves 4–6) of tea saplings (A3) and the metabolic rate of glyphosate metabolism into AMPA (B3).

between the cleanup methods, the standard solutions of glyphosate and AMPA were used rather than matrix match calibration (Figure S1). The lowest recovery of both glyphosate and AMPA resulted from using HLB only. This MS signal decreased possibly because the matrix effect was higher than the other two methods. When the leaf extract was re-extracted with CH₂Cl₂, the recoveries of glyphosate and AMPA increased. The highest recoveries occurred when the re-extracted sample was mixed with PVPP and GCB. This QuEChERS method resulted in 84.2% and 72.3% recovery rates for glyphosate and AMPA, respectively. It was encouraging to see that the best recovery rate was achieved with the quickest and least expensive sample preparation method.

Method Validation. The standards ranged from 5 to 500 µg L⁻¹ as described in the LC–MS/MS Analysis section. The peak areas for each standard concentration were plotted and fit to a linear equation, from which the correlation coefficients of the two compounds were obtained. As shown in Table S1, the linearities of the standard curves of the two compounds were good, and the *r*² values were higher than 0.999.

The recovery rates and relative standard deviation (RSD) of the two compounds from the roots, stems, and leaves of tea

plant are as shown in Table 1. The recovery rate of glyphosate ranged from 82.3 to 116.0% and of AMPA from 72.3 to 94.6%. The RSD (*n* = 6) values were 4.70–12.99% and 6.50–13.43%, respectively. The recovery rate and RSD values meet the requirement for pesticide analysis. The LOQ of glyphosate was 0.1 mg kg⁻¹ for both glyphosate and AMPA in leaf samples and 0.05 mg kg⁻¹ for stem and root samples.

Phytotoxicity of Glyphosate to Tea Plant. To investigate the phytotoxicity of glyphosate to tea plants, tea saplings were cultivated in a nutrient solution containing different concentrations of glyphosate (0, 5, 50, 200, or 2000 mg L⁻¹). The tea leaves were observed at different times (from 0 to 21 days; Figure 4). When grown in 2000 mg L⁻¹ glyphosate, mature leaves on 7 DAT and young leaves on 8 DAT showed some dark brown spots. This browning gradually spread over the entire leaves, which began to fall off on 14 DAT. For tea plants cultivated in 200 mg L⁻¹ glyphosate, the mature leaves began to develop dark brown spots on 8 DAT, and the young leaves began to appear dark brown on 10 DAT. The browning increased gradually and the leaves fell off eventually. This negative effect can come from inhibited acquisition of micronutrients such as Mn, Zn, Fe, and B, which

are involved in plant disease resistance mechanisms.^{13,41} However, with the concentrations of 5 mg L⁻¹ and 50 mg L⁻¹ glyphosate, the tea leaves did not show any phytotoxicity over 14 days and showed no significant difference to the control sample. These data indicate that the application of glyphosate needs to be controlled under 50 mg L⁻¹ to avoid toxicity to the tea plants.

Uptake, Transport, Metabolism, and Distribution of Glyphosate in Tea Plant. Afterward, the tea saplings were treated with water nutrient solution containing 5 mg L⁻¹ glyphosate for 21 days in blue box. The uptake, transport, metabolism, and distribution of glyphosate and AMPA in tea plants were investigated. Results shown in Figure SA1 indicate that the amount of glyphosate in the roots increased gradually with time, from 113.54 mg kg⁻¹ on day 0 (2 h) to 294.87 mg kg⁻¹ on day 5, which marked the highest accumulation level, and then decreased to 66.94 mg kg⁻¹ on day 21. Compared to the concentration of glyphosate (5 mg L⁻¹) in the nutrient solution, the accumulation coefficient of roots was about 3.5 mg at 2 h. The amount of glyphosate in stems increased from 17.80 mg kg⁻¹ on day 0 to 46.83 mg kg⁻¹ on day 5 and then decreased to 12.78 mg kg⁻¹ on day 21. The amount of glyphosate in mature tea leaves increased from 0.35 mg kg⁻¹ on day 0 to 14.49 mg kg⁻¹ on day 5 and then decreased to 7.07 mg kg⁻¹ on day 21. The amount of glyphosate in young leaves increased from 0 mg kg⁻¹ on day 0 (after 2 h) to 13.84 mg kg⁻¹ on day 5 and then decreased to 2.17 mg kg⁻¹ on day 21. After glyphosate absorption by the roots, the young leaves accumulated 1.65 mg kg⁻¹ glyphosate on the first day. This indicated that glyphosate was transferred from roots to leaves through transpiration pull.⁴² The cumulative amount of glyphosate in each part of the tea sapling decreased gradually from the fifth day because glyphosate was gradually degraded to AMPA and other metabolites. On days 1, 3, 7, 10, 14 and 21, the cumulative amounts of glyphosate in the young leaves (leaves 1–3) were 1.65, 2.61, 3.55, 3.57, 3.86, and 2.71 mg kg⁻¹, respectively, amounts that were all less than half of the amounts in mature leaves (4.87, 8.09, 12.78, 10.63, 17.31, and 7.07 mg kg⁻¹, respectively) (Figure 5A3). Interestingly, the glyphosate level in young and mature leaves was almost equal on day 5 (13.84 and 14.49 mg kg⁻¹). From these results, it seems that teas prepared with young leaves would not only be of higher grade, but also would also have lower glyphosate content, making these products relatively safer than those made from mature leaves.

The levels of the glyphosate metabolite AMPA were determined in roots, stems, and leaves of tea saplings (Figure 5A2). AMPA was not detected in the young or mature leaves from day 0 to 21 but did increase in stems and roots with time (in roots, from 0 to 2.76 mg kg⁻¹ and in stems from 0 to 0.53 mg kg⁻¹). The absorption of glyphosate and the production of AMPA in the whole plant was also quantified (Figure 5B1,B2). The cumulative amount of glyphosate in the plant as a whole ranged from 67.70–133.99 mg kg⁻¹ from 0 to 3 days, remained relatively constant during days 3 to 10, and then decreased to 25.26 mg kg⁻¹ by day 21. Meanwhile, the AMPA increased gradually from 0 to 1.58 mg kg⁻¹ from 0 to 21 days. The metabolic rate of glyphosate transformation into AMPA in whole plant was calculated (Figure 5B3). The rate of glyphosate metabolism into AMPA increased from 0.19% on the 3rd day to 5.89% on the 21st day. AMPA represented a very small portion of the PMG/AMPA pool, and so had a lesser impact on tea or tea product safety than did glyphosate.

This report represents the first investigation into the uptake, transport, metabolism, and distribution of a systemic pesticide in tea plants over time. This study informs growers that by controlling the free glyphosate residues in the soil, one can produce tea product free of glyphosate. It also reminds researchers and growers alike that systemic pesticides such as glyphosate can transfer through the root to leaf. This represents another route generating potential exposure to consumers that differs from the direct contact of pesticide residues applied to edible parts.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.jafc.7b02474.

Optimization procedure for different cleanup methods and linear equations, matrix effect (ME), recoveries of our proposal method (PDF)

■ AUTHOR INFORMATION

Corresponding Author

*E-mail: hry@ahau.edu.cn. Phone: +86–551-65786401.

ORCID

Ruyan Hou: 0000-0003-4423-694X

Author Contributions

§M.T. and W.G. contributed equally to this work.

Notes

The authors declare no competing financial interest.

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